Understanding Air Quality Through Data

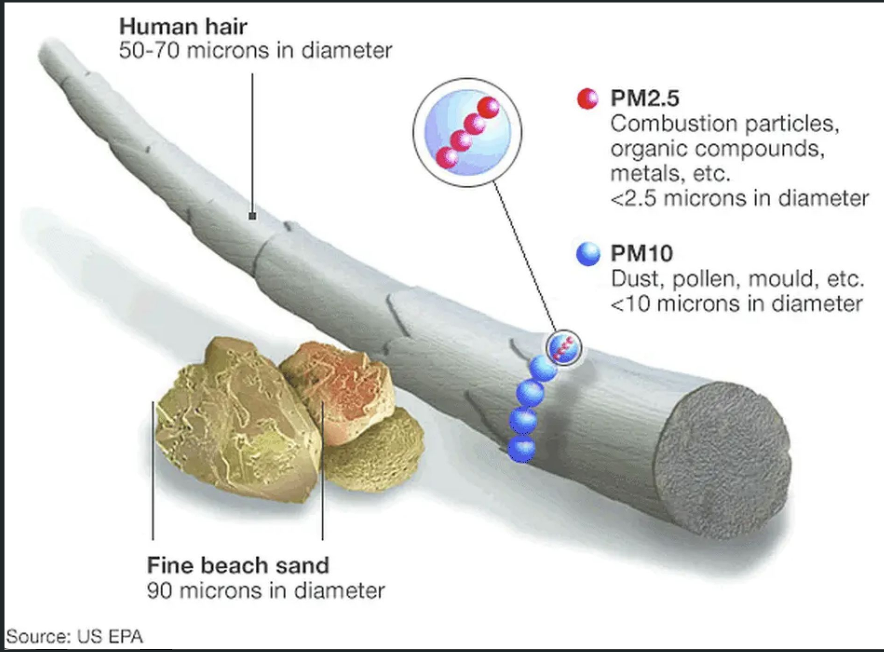
Justin Peabody, Dr. Whalen

The CLEAR lab

- CLEAR = Community and Local Engaged Air Research
- Multidisciplinary student/faculty air quality research team
- Goals:
  - air quality (AQ) sensor installation
  - analyzing/mapping sensor data
  - research and public education

PM2.5

- Particulate matter: suspended air particles
- PM2.5: particulate matter < 2.5 um in diameter
- Smaller particles pose highest health risk
- Sources: fossil fuels, industry, vehicle exhaust, wildfire smoke



PurpleAir Sensors

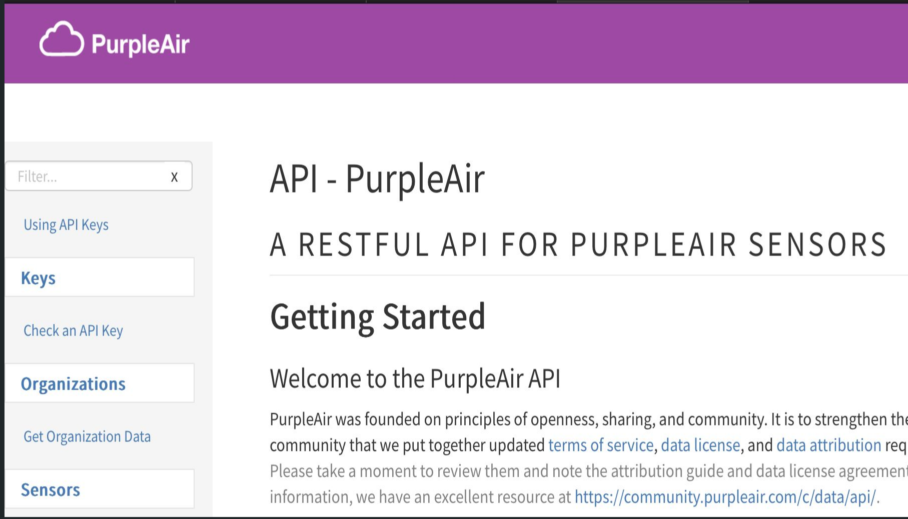
- PurpleAir: affordable AQ sensor manufacturing company
- Offers smaller, more portable, higher-granularity sensors than EPA
- 92 sensors in Chicago, 62 of which are active

PurpleAir Sensors

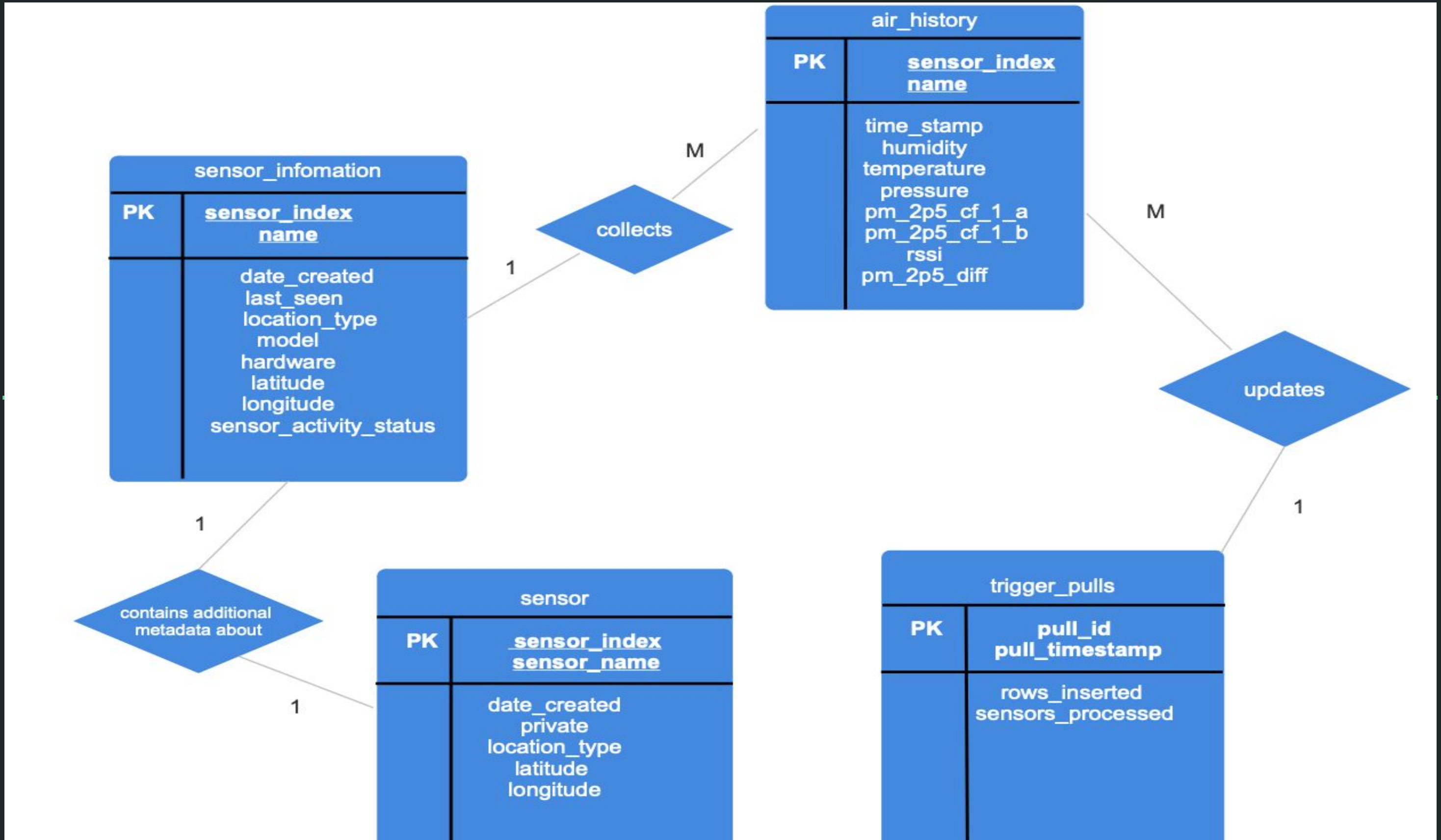
- work both indoors and outdoors
- Measure PM2.5, humidity, temperature, and atmospheric pressure
- Readings uploaded to cloud based server, accessible via API

PurpleAir API

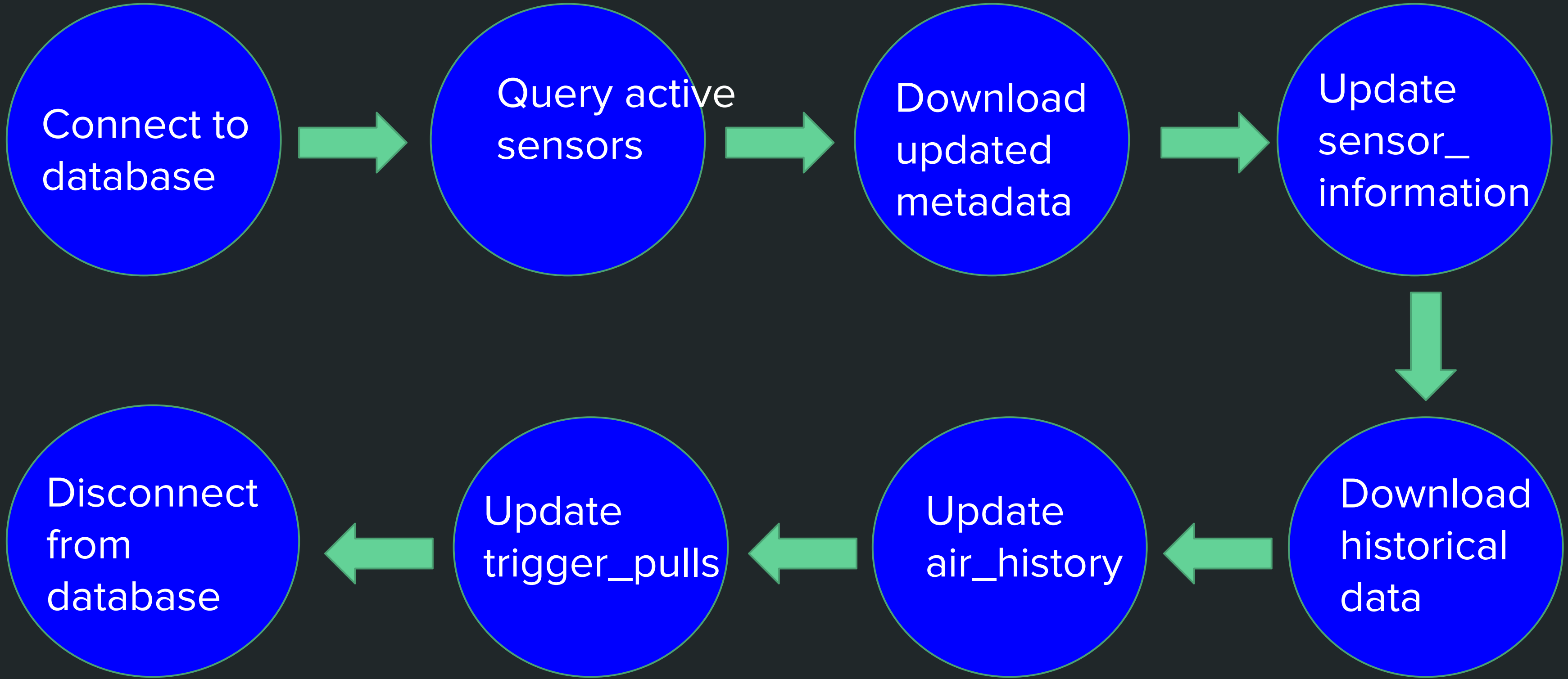
- API = Application Programming Interface
- Allows connection between servers/applications



Database Structure



Database update script



Creating a trigger

- Trigger=event or condition that “triggers” code to automatically run

- Eliminates the need for manual execution

```
> <?xml version="1.0" encoding="UTF-8"?>
> <!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://apple.com">
> <plist version="1.0">
> <dict>
>     <key>Label</key>
>     <string>com.user.airquality</string>
>     <key>ProgramArguments</key>
>     <array>
>         <string>/Library/Frameworks/R.framework/Resources/bin/Rscript</string>
>         <string>/Users/Shared/USRE_Air_Quality_project/24_hr_update_script.R</string>
>     </array>
>     <key>EnvironmentVariables</key>
>     <dict>
>         <key>PATH</key>
>         <string>/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin:/opt/homebrew/bin</string>
>     </dict>
>     <key>StartCalendarInterval</key>
>     <dict>
>         <key>Hour</key>
>         <integer>0</integer>
>         <key>Minute</key>
>         <integer>0</integer>
>     </dict>
>     <key>StandardOutPath</key>
>     <string>/Users/Shared/USRE_Air_Quality_project/run.log</string>
>     <key>StandardErrorPath</key>
>     <string>/Users/Shared/USRE_Air_Quality_project/run.log</string>
> </dict>
> </plist>
> EOF
```

Future goals

- Create databases for sensors from other companies (AirGradient, Clarity)
- Continued sensor installation
- Create a public sensor dashboard

What I learned

- Introduction to SQL programming
- Queries and database operations
- ER diagrams for databases

Database update script